

GPA7XXXA Release Note

V1.3.1

September 18, 2025

1. Note

- G+IDE for ARM v1.2.0 can be obtained through the following URL.
https://www.generalplus.com/1LVlangLNxxSVyySNservice_n_support_d

2. Driver

- File System:
 - (1) Fixed the issue with writing content to an existing file.
 - (2) Fixed the issue with incorrect MBR check.
- GPU:
 - (1) Added GLLite driver and remove old GAL3D driver.
 - (2) Update GPU demo and 3D avatar demo to GLLite.
- JPEG:
 - (1) Update image size setting and MCU block number calculation.
- MultiMediaParser:
 - (1) Update loop play function.
- NAND:
 - (1) Update lib_nand_gpa7XXXA_g+ide_freertos.a. These lib support NAND/SPINAND UUID AES encryption to bin file, new NF_APP_PART_INFO_EXT structure, reading LC API NandLCRead, NandLCGet, refer drv_l2_nand.h. For code demo, refer nand_console_demo.c command "nandemo app updatefw", "nandemo app partinfo", "nandemo app get_lc_body", "nandemo app get_lc_fuse".
 - (2) Fixed NandAppSycMappable bug that PutSwapPhysicBlock fail to release map block, then cause GetMapTblInfo fail and print NP Failed after swap block less than 1. Note that this issue also requires update CodePacker 1.9.9.22's ISP/BootLoader to fix this bug.
- SD/eMMC:
 - (1) Fixed CMD12 may send one more time when eMMC.

- SPINAND:
 - (1) Update lib_nand_spi_gpa7XXXA_g+ide_freertos.a. These lib support NAND/SPINAND UUID AES encryption to bin file, new NF_APP_PART_INFO_EXT structure, reading LC API NandLCRead, NandLCGet, refer drv_l2_nand.h. For code demo, refer nand_console_demo.c command "nandemo app updatefw", "nandemo app partinfo", "nandemo app get_lc_body", "nandemo app get_lc_fuse".
 - (2) Fixed NandAppSyncMaptable bug that PutSwapPhysicBlock fail to release map block, then cause GetMapTblInfo fail and print NP Failed after swap block less than 1. Note that this issue also requires update CodePacker 1.9.9.22's ISP/BootLoade to fix this bug.

3. Demo

- 3D avatar demo:
 - (1) Add gc1084 support.
 - (2) Add 4:3 FD image patch for FD quality.
 - (3) Add draw ROI on display.
 - (4) Fixed color format issue.
- LVGL demo:
 - (1) Update the method of determining the width and height of the LVGL player widget.
 - (2) Fixed the issue that occurs when creating multiple LVGL player widgets.
- MP3 decode NVRAM demo:
 - (1) Added support for eMMC resource.
 - (2) Fixed potential read 0 sector issue.
 - (3) Fixed GPA7003A_MP3Dec_NVRAM_Demo_SPIFC_release play mp3 no sound
- NAND SD SPINAND MSDC Demo:
 - (1) emmc_console_demo.c: Added emmcdemo get_lc_body comamnd.
 - (2) spi_console_demo.c: Added spidemo get_lc_body command
 - (3) nand_console_demo.c: Added get_lc_fuse, get_lc_body command.
 - (4) nandemo app updatefw console command support two code part update, update verify compare with original bin file.
 - (5) nandemo app updatefw console command support update code with UUID AES encryption. Note that to boot UUID AES encrypted bin, have to use G+ Code Packer V1.10.0.26, and select Protection.
 - (6) nand_update_app_partiton_demo.c: When using CodePacker 1.9.9.22 and later, when Checksum or Protection is on, if want calc_checksum value want same to CodePacker 1.9.9.21 and previous version, have to COMPATIBLE_WITH_CODEPACKER_19921 to 1.

(7) Fixed nand console command emmcstop not working when drv is 2(SD controller 0), cause is SD module and clock have been off at bootloader.

- Video recorder demo:

(1) Added handling of audio missing in video recorder.

- New Demo/Sample code:

(1) WebRTC demo.

V1.3.0

April 28, 2025

1. Note

- Getting Started with GPA7 Series please refer to following video.
<https://www.youtube.com/watch?v=V5jBh7lYRK0>
- G+IDE for ARM v1.2.0 can be obtained through the following URL.
https://www.generalplus.com/1LVlangLNxxSVyySNservice_n_support_d
- GPA7XXXA V1.3.0 release based on SVN revision 18107.

2. Driver

- Clock:
 - (1) Enabled the GP420 converter clock to fix the hang-up issue caused by the Cortex-A7 issuing an unexpected access to address 0x60000000.
- CDSP:
 - (1) Added ISP auto contrast function (UVC demo default disable).
 - (2) Fixed the issue that ISP AE is unstable in some environments.
 - (3) Fixed 3DNR system FIFO address setting bug.
- Display:
 - (1) Added TFT2 scale up and edge function.
 - (2) Fixed DISDEV_TFT2_HDMI576P_50HZ setting.
- Ethernet:
 - (1) Fixed "Eth Tx is busy" issue.
- File System:
 - (1) Added support for exFAT. Set FS_EXFAT_SUPPORT in gplib/fs/fs.h to 0 to disable exFAT or to 1 to enable it.
- GPU:
 - (1) Fixed GPU memory cache issue.
- I2C:
 - (1) Fixed I2C slave mode (GP328xxxA as master) Rx/Tx transfer error in non-DMA/DMA operation mode.

- JPEG:
 - (1) Added support for YUV400.
- MultiMediaParser:
 - (1) Added loop play function.
 - (2) Added decrypt function call.
 - (3) Added memory management API register functions.
- NAND:
 - (1) Added support for SkyHigh parallel NAND 1G bit, 2G bit part number.
Note: version of G+ Code Packer suggest use V1.9.9.21 and later.
- SD:
 - (1) Added detecting CRC error in SD3.0 read/write DMA timeout loop to fix card out problem.
 - (2) Fixed SD3.0 pinmux using SD2.0 driver.
 - (3) Fixed error clock setting for SD1 pinmux not using 1.8v.
 - (4) Fixed SD 3.0 controller compatible with SD Card issue.
 - (5) Fixed SD 3.0 controller when busy reading SD Card condition sometime happen time out.
- Sensor:
 - (1) Added GC1084 driver.
 - (2) Added GC2093 driver.
 - (3) Added GC8613 driver.
 - (4) Added IMX335 driver.
 - (5) Added sensor stream on/off function.
- SPIFC:
 - (1) Added support for some SPI part numbers of XTX, ESMT, PUYA, INFINEON, WINBOND, MXIC, MICRON, GIGADEVICE.
Note: for booting mode, ISP mass production mode, version of G+ Code Packer suggest use V1.9.9.21 and later.
- SPINAND:
 - (1) Fixed lib_nand_spi_gpa7XXxA_g+ide_freertos.a bug.
Note: For new part numbers support such as XTX series and more other part numbers, version of G+ Code Packer suggest use V1.9.9.20 and later.

- SPINOR:
 - (1) Added support for some SPI part numbers of XTX, ESMT, PUYA, INFINEON, WINBOND, MXIC, MICRON, GIGADEVICE.
 - Note: for booting mode, ISP mass production mode, version of G+ Code Packer suggest use V1.9.9.21 and later.
- System:
 - (1) Added bus priority setting.
- UART:
 - (1) Fixed UART baud rate setting bug.
- USB D:
 - (1) Added master1 controller interrupt handling.
 - (2) Fixed remote-wakeup feature mechanism.
- USB H:
 - (1) Added software watch-dog for ISO packet transfer.
 - (2) Fixed OHCI driver the next iTD of the last iTD overlayed.
 - (3) Fixed OHCI driver for plugin/pluginout.

3. Demo

- Audio decode demo:
 - (1) Fixed MSG_AUD_DMA_DBF_START is not sent when audio queue not full.
 - (2) Fixed video decoder playback issues with videos containing minimal audio data.
- Camera auto framing demo:
 - (1) Fixed UVC mode issue.
- Cartoonization demo:
 - (1) Added Animated East/Animated West/Aging/Kids type.
- Digital video demo:
 - (1) Fixed video decoder playback issues with videos containing minimal audio data.
- GPU demo:
 - (1) Added support for GL_VIV_direct_texture.

- Image encode demo:
 - (1) Added Gray Scale Mode.
- Image decode demo:
 - (1) Added Gray Scale Mode.
- MP3 decode NVRAM demo:
 - (1) Added GPA7xxxA_MP3Dec_NVRAM_Demo_SPINAND_release.gpcmprj, it plays mp3 song resource that at SPINAND resource part.
 - (2) Added GPA7003A_MP3Dec_NVRAM_Demo_SPIFC_release.gpcmprj, it plays mp3 song resource but code is running on SPI flash.
- NAND SD SPINAND MSDC demo:
 - (1) Added spidemo command console to GPA7xxxA_sd_spinor_msdc_release project. At cmd> prompt typing "spidemo" for list of command.
 Note: version of G+ Code Packer suggest use V1.9.9.21 and later.
 Note: Product that access LC's fuse suggests using G+ Code Packer 1.10.0.26 and later version.
 - (2) Added emmcdemo command console to GPA7xxxA_sd_nand_msdc_release project. At cmd> prompt typing "emmcdemo" for list of command.
 Note: version of G+ Code Packer suggest use V1.9.9.21 and later.
 Note: Product that access LC's fuse suggest using G+ Code Packer 1.10.0.26 and later version
 - (3) Added GPA7003A_sd_spinand_msdc_release_SPIFC.gpcmprj, it access SPINAND but code running on SPI flash.
- QRCode/BarCode demo:
 - (1) Added support for IMX415 camera module.
 - (2) Added support for drawing string to show QRcode/BarCode result on display.
- USB D CDC demo:
 - (1) Added USB D detection IO define in application_cfg.h.
 - (2) Fixed USB2.0CV fail.
- USB D HID demo:
 - (1) Fixed configuration descriptor length for hid raw mode.
- USB D MSDC demo:
 - (1) Added USB D detection IO define in application_cfg.h.

- USB D UAC demo:
 - (1) Added USB D detection IO define in application_cfg.h.
 - (2) Fixed up full speed enumeration bug.
 - (3) Fixed USB2.0CV fail by modifying configuration descriptor.
- USB D UVC demo:
 - (1) Added Y800 format.
 - (2) Added UVC MJPEG encoder input format define in application_cfg.h to select GP420 or YUYV.
 - (3) Added UVC MJPEG encoder output format define in application_cfg.h to select YUV420 or YUV422 when the input format is GP420.
 - (4) Added UVC video controls.
 - (5) Added USB D detection IO define in application_cfg.h.
 - (6) Fixed ISO in by DMA size over 4MB bug.
 - (7) Fixed pause and resume bug when previewing.
 - (8) Fixed previewing bug in Gucvview.
 - (9) Fixed USB2CV fail.
- USBH HID demo:
 - (1) Fixed memory leakage while task exiting then creating.
- USBH MSDC demo:
 - (1) Fixed U3 flash disc bus disconnected while USB host MSDC enumeration.
- USBH UVC demo:
 - (1) Added obtaining reverse parking status function.
 - (2) Added monitoring USB host OHCI isochronous stream status for bus error.
 - (3) Added streaming format auto-selection (JPEG+YUV->JPEG, YUV only->YUV).
 - (4) Added 4-port hub supporting, only for HS device.
 - (5) Fixed the hanging up problem while device removing then inserted.
- Video recorder demo:
 - (1) Added loop recording.
 - (2) Fixed video recorder file size reach limit not stop issue.
 - (3) Fixed JPEG encode error.
- New Demo/Sample code:
 - (1) Added AEC demo.

- (2) Added CAN sample.
- (3) Added DMS demo.
- (4) Added dual camera demo.
- (5) Added face identify demo.
- (6) Added Gesture detection demo.
- (7) Added GPDLA NN examples.
- (8) Added GPU 3D avatar demo.
- (9) Added HTTP Client Demo.
- (10) Added IRAM sample.
- (11) Added license plate recognition.
- (12) Added LVGL demo.
- (13) Added RCTA demo.
- (14) Added RTSP demo.
- (15) Added segmentation demo.
- (16) Added smile detection demo.
- (17) Added SPI slave sample.
- (18) Added USB D RNDIS demo.
- (19) Added voice recognition demo (DSpotter32).

V1.2

April 28, 2023

1. Note

- All of the demo/sample code project choose GCC version 11.3.1 to compile in this platform code. Please use G+IDE for ARM v1.1.3 which support GCC version 11.3.1 to open demo/sample code project.
- G+IDE for ARM v1.1.3 can be obtained through the following URL.
https://www.generalplus.com/1LVlangLNxxSVyySNservice_n_support_d
- GPA7XXXA V1.2 release based on SVN revision 15826.

2. Driver

- Crypto:
 - (1) Disable auto DMA in drv_l1_hash_init().
- GPU:
 - (1) Update GPU driver (libGAL.a) to avoid unknown shader code issue.
- H.264:
 - (1) Add soft reset before decode each frame to fix potential H.264 engine state incorrect issue.
- JPEG:
 - (1) Fixed JPEG decode output image size overwrite issue.
- SD:
 - (1) Improve no card case, drv_l2_sd_init quit quickly.
 - (2) drv_l2_sd_init time out after 2 second when bad SD card could not feedback voltage is ok.
- SPINOR:
 - (1) Support Winbond, Micron, XTX, GigaDeice 2 bit, 4 bit bus width, and 256M bit spi flash.

3. Demo

- USB CDC demo:
 - (1) Fixed EP 1/2 bulk out NAK problem as CDC EP = EP 1/2.
- USB MSDC demo:
 - (1) Add more NAND partitions for multi-LUN device.

- USB D UVC demo:
 - (1) Fixed USB D UVC no toggle frame ID bug for YUY2 format.
- USB H UVC demo:
 - (1) Fixed setup-command error for long USB cable by retrying 5 times
 - (2) Fixed OHCI plug-out port change signal triggering EHCI port change interrupt while OHCI exiting and EHCI takes over it.
 - (3) Fixed display fragment while adjusting IQ parameters.
 - (4) Add UVC reversing parking detection function.

V1.1

March 31, 2023

1. Driver

- Sensor:
 - (1) Support SC4336.
 - (2) Support OV9281.
 - (3) Fixed sensor MCLK 27M/37M125 outputs failed.
- Panel:
 - (1) Support mipi_display_ICN6202LVDS-CF0380D (1024x768).
 - (2) Support mipi_display_FX070-DHM18 (1024x600).
- FIQ:
 - (1) Change IRQ, FIQ priority to fix critical section not block FIQ issue.
 - (2) Disable UART FIQ by default.
- H.264:
 - (1) Update H.264 & scan2mb reset.
 - (2) Rename H.264 encode parameter "fifo_mode" to "out_fifo".
 - (3) Update H.264 driver comments and document for 16 alignment requirement.
 - (4) Fixed FBC error handling, and add FBC ratio as parameter.
 - (5) Fixed H.264 notify bug in multi-instance.
- MultiMediaParser:
 - (1) Fixed data overflow cause jpeg decode fail.
 - (2) Read offset in 64 byte align.
- File System:
 - (1) Fixed FAT12 fat chain happen cross link error.
 - (2) Fixed mount 64G SD card exFAT fail.
- CSI:
 - (1) Add CLK_EN1_PPU_MUX clock setting.
 - (2) Correct CSI long burst setting.

- Object Detect:
 - (1) Modify scaler crop flow.
- PPU:
 - (1) Modify PPU DMA flow.
- HDMI:
 - (1) HDMI exit flow.
 - (2) HDMI 720p/1080p "fast_mode_enable" flow.
- DLA:
 - (1) Add pose / hand module flow.
- USBBD:
 - (1) Modify the used timer for USB detection to the same one with tiny counter.
- USBH:
 - (1) Fixed EHCI driver error handling.
 - (2) Define each device class in project setting.
 - (3) Constraints some UVC class headers and functions.
- New Driver:
 - (1) Temperature driver.

2. Demo

- MP3 decode NVRAM demo:
 - (1) Add playing MP3 from NAND APP Area's NVRAM resource.
- G+ ISP tool:
 - (1) Fixed capture/preview failed.
- Change system clock demo:
 - (1) Add a new API "INT32S drv_l2_sys_cpu_clk_set(int systemclk, CPU_SYS_RATIO csratio)".
User can call this API when the system switches the CPU and system frequency and does not need to switch the DDR frequency.
 - (2) Modify 32K and 24M mode return flow.

Note: For detailed usage, please refer to the demo documentation.

- Sys recover demo:
 - (1) Add CPU registers recover after system recovery.
- Wi-Fi demo:
 - (1) Support using RTSP to transmit Audio (MULAW).
- USB D UVC demo:
 - (1) Support GL3004 FW update.
 - (2) Remove unused log for GL3004 extension unit.
 - (3) Fixed auto mode stripe problem caused by the CDSP reset in CDSP IRQ handler.
 - (4) Modify exiting flow for auto mode to 2 channels.
 - (5) Add GC0308 sensor.
 - (6) Modify USB D UVC configuration descriptor to remove still image descriptor and add color matching descriptor for MJPEG and H.264 format.
 - (7) Fixed auto mode 2 channel plug-out to in hang up bug.
 - (8) Fixed guvcview not working problem.
 - (9) Modify configuration descriptor to disable unsupported controls.
- Image decode demo:
 - (1) Modify function "jpeg_file_decode_output()" to check if the input image is progressive jpeg and then return fail to enter the "jpeg_file_decode_and_scale()" function to decode progressive jpeg.
- Tracking camera demo:
 - (1) Fixed tracking camera demo exit may block issue.
 - (2) Add more head space.
 - (3) Support system FIFO.
 - (4) Support IMX415.
 - (5) Fixed exit hang in UVC mode.
- Digital video demo:
 - (1) Add H.264 reset when H.264 FIFO catch buffer overflow 3 times.
 - (2) Fixed video packer task exit timeout problem after the video file reaches the target size.
- Video recorder demo:
 - (1) Disable DN3D if "sensor_id" > 0 (dual sensor).

- Scaler demo:
 - (1) Add crop flow for scaler.
- FD demo:
 - (1) Modify system FIFO flow.
 - (2) Modify PPU draw flow.
 - (3) Fixed DRAW_FRAME_RATE_WITH_SENSOR_EN = 1 compiler error.
- Body demo:
 - (1) Modify system FIFO flow.
- Pose demo:
 - (1) Modify array "send_audio[2]" to variable "send_audio".
- USBH UVC demo:
 - (1) Fixed problems while USBH UVC adjusting IQ parameters.
 - (2) Process device inserting/removing to/from UVC host heap memory management.
- USBH MSDC demo:
 - (1) Register initial operation table for USBH MSDC demo.
- GPU:
 - (1) Update GPU demo to support HDMI display
- Video Recorder Demo:
 - (1) Fixed memory leak.
 - (2) Fixed audio noise.
 - (3) Fixed CSI sensor restart fail issue.
 - (4) Use separate JPEG module for dual MJPEG stream recording to improve performance.
- H.264 sample:
 - (1) Clear input parameter structure to zero to avoid unexpected error.
- New Demo/Sample code:
 - (1) HDMI demo.
 - (2) Hand demo.
 - (3) USBH CDC demo.
 - (4) USBH HID keyboard demo.

(5) Temperature demo.

3. Misc.

- Add GPA7xxxA_pmic_gdb folder to support control GP GPY02XXX PMIC through ICE mode.
For better effect, have to use G+IDE V1.1.2.2 and later version.

4. Note

- GPA7XXXA V1.1 release based on SVN revision 15676.